Business Requirements Document

NYC Ride-Hailing, Weather & EV Charging Access

Course	DL3004 Data Warehousing & BI Lab, Fall 2026, FAST-NU Lahore
Instructors	Mr. Muhammad Ishaq Raza (Course), Mr. Shahzeb Mubashar (Lab)
Team	Ayza Ahmed - 24L-2577, Eman Adil - 24L-2589, Dua Sarfaraz - 24L-2587, Gul-e-Zara - 24L-2592, Rumesha Naveed - 24L-2603
Phase	1 - Business Analysis & Dimensional Design

1. Business Problem

Business problem in one line:

We help ride-hailing planners in New York City see how rain and nearby EV chargers affect trips, waiting time and driver pay in each zone. Today this information sits in separate datasets, so nobody sees the full picture.

2. Stakeholders and What They Need

Stakeholder	What they need
Fleet operations manager	Know which zones and hours are busiest and how rain changes demand and driver pay.
City transport planner	See which busy zones are far from chargers.
Charging network planner	Decide where new chargers would help most.

3. Scope

In scope	Out of scope
Uber and Lyft trips (TLC High Volume data), 500k-row sample; hourly rainfall for NYC; public EV charging stations in NYC	Yellow, green and other taxi types; live data; battery or vehicle-level data; demand prediction

4. Business Requirements

ID	Requirement	Priority
BR-01	Store every sampled trip with pickup zone, drop-off zone, time, distance, fare, tip and driver pay.	High
BR-02	Calculate customer waiting time (pickup time minus request time).	High
BR-03	Link each trip hour to hourly rainfall.	High
BR-04	Group rainfall into levels (none, light, heavy).	Medium
BR-05	Place every charging station inside a taxi zone.	High
BR-06	Keep history when a station changes its network, ports or status (SCD Type 2).	High
BR-07	Analyse time by year, quarter, month and day, and place by borough and zone.	High
BR-08	Reject bad records, record the reason, and reconcile source rows with warehouse rows.	High
BR-09	Show results in a Power BI dashboard with maps and OLAP views.	Medium

5. Data Sources

Source	Type	Used for
NYC TLC High Volume FHV trips	PostgreSQL	Trips, waiting time, fares, driver pay
Open-Meteo historical weather	REST API	Hourly rainfall
AFDC / NREL Alternative Fuel Stations	JSON (API)	EV charging stations
NYC Taxi Zone lookup and polygons	Reference CSV	Zone names, boroughs, map shapes

6. Assumptions and Success Criteria

- All data is real and public. No data is simulated.
- The trip data does not show which vehicles are electric, so we study ride-hailing as a whole and link it to charger access by zone. It has no surge multiplier, so driver pay per mile is used as a simple price signal.
- Stations only have latitude and longitude, so we match them to zones with the zone polygons.
- **Success:** the warehouse answers all 10 business questions, row counts match between source and warehouse with rejects logged, and the dashboard shows rain impact and charger gaps on a map.